



Graphs in Machine Learning

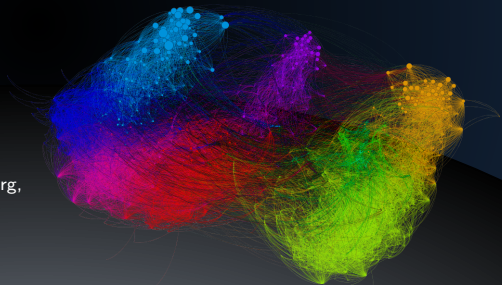
Movie Recommendations

Graph Distance Approaches

Michal Valko

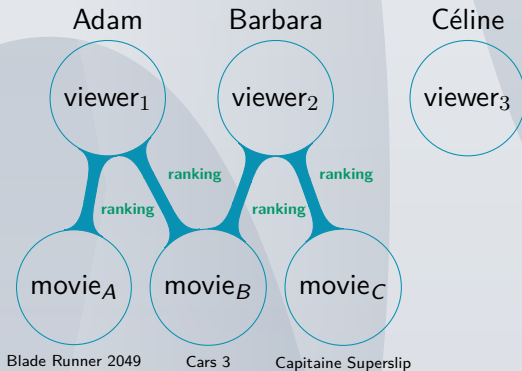
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Partially based on material by: Ulrike von Luxburg,
Gary Miller, Doyle & Schnell, Daniel Spielman



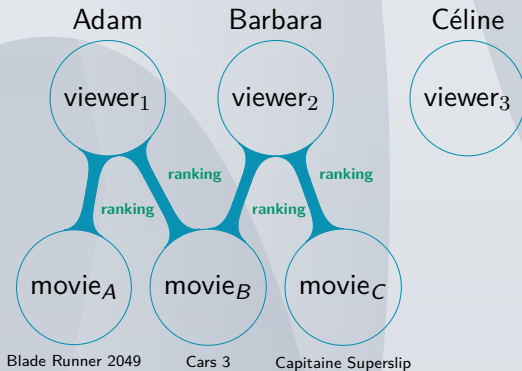
Use of Laplacians: Movie recommendation

How to do movie recommendation on a bipartite graph?



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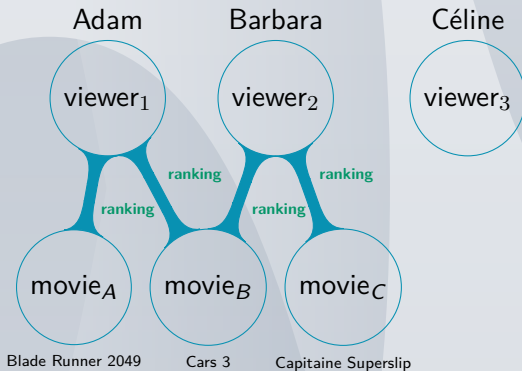
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Question: *Do we recommend Capitaine Superslip to Adam?*

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Use of Laplacians: Movie recommendation

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Use of Laplacians: Movie recommendation

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Idea₁: maximally weighted path

$$\text{score}(v, m) = \max_{vPm} \text{weight}(P) = \max_{vPm} \sum_{e \in P} \text{ranking}(e)$$

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<https://misovalko.github.io/mva-ml-graphs.html>